Sat Arora

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US & Canadian Dual Citizen – No work permit required

Education

University of Waterloo GPA: **91.4%**
Bachelor of Computer Science (Artificial Intelligence Specialization) *Sep. 2021 – Apr. 2026*

- Awarded **1 of 10** René Descartes National Scholarships **($20000)** from the Waterloo Faculty of Mathematics
- President of the Waterloo Computer Science Club; cofounded events, initiatives, and the CS Class Profile

Experience

(Incoming) Software Engineer Intern Sep. – Dec. 2025
Ramp *New York, NY*

Software Engineer Intern May – Aug. 2025
Snowflake *Menlo Park, CA*

- **Architected an insight system** to alert customers when specific SQL clauses in workloads incurred extra compute due to unnecessary duplicate filtering, optimizing performance for **10%** of all queries
- Suggested warehouse modifications to users when queries were queued due to warehouse overload of **10+ minutes**

Software Engineer Intern Sep. – Dec. 2024
Ramp *New York, NY*

- Enhanced merchant identification accuracy by **70%** through an **embedding vector centroid search of receipt OCR data**, with additional LLM queries for **auto-classification** of merchant categories
- Spearheaded the integration of **WhatsApp authentication** into authentication flows, enabling a projected **1,000+ customers** per month to adopt MFA
- Led the backend overhaul of mileage reimbursements, adding live GPS tracking, auto-generated receipts, deduction of personal commutes, and the option to pin and use frequent trips – see more on this blog post!

Infrastructure Engineer Intern May – Aug. 2024
Matroid *Palo Alto, CA*

- Accelerated Docker container startup time by **50%** by removing redundant pre-loaded models and optimizing computer vision code in Python, resulting in faster deployment with leaner dependencies
- **Engineered a real-time device detection system** that enables on-premise compute systems to automatically discover and utilize devices for live monitoring, enabling **real-time inference pipelines**

Software Engineer Intern May – Aug. 2023
Miovision *Kitchener, ON*

- Implemented REST endpoints to configure traffic detection devices with a new Advanced Transportation Controller (ATC) standard using **Java** and **SQL**, deploying to **3000+ intersections in North America**

Data Engineer Intern Sep. – Dec. 2022
Ontario Ministry of Finance *Toronto, ON*

- Reduced database migration time to a **SQL** server solution by automating the extraction of data from old databases, resulting in **90%** faster total extraction time and **saving over 1 month** in the migration process

Projects

Class Activation Mapping and Object Detector | *Python, PyTorch, Matplotlib, NumPy*

- Trained a face-detection **CNN** using 2000+ images and tested with 500+ images, achieving **99%+** accuracy
- Extracted **image heat maps** for classes using **Global Average Pooling** on the CNN's final convolutional layer, which identified the location of dominant facial features with high accuracy, including on multiclass classifiers

Multicore & Parallel Algorithms | *C++*

- Implemented expanding hash tables, lock-free queues, external & internal binary search trees, and multi-word compare-and-swap (KCAS) for concurrent doubly linked lists, achieving **near-linear scaling to 16 threads**
- Leveraged compiler profiling tools such as **perf stat** and **thread pinning** to reason consistency across CPU cores

Technical Skills

Languages: Python, C/C++, Java, JavaScript/TypeScript, Bash, SQL, C#, .NET, R, Racket
Technologies: Node.js, AWS S3, PyTorch, NumPy, Pandas, Spark, CUDA, Linux, Kubernetes, MongoDB, React.js